

Problem Statement

In this project, you will analyze an online retail dataset containing customer transactions, product details, quantities, prices, and purchase dates.

Businesses use such data to understand sales trends, customer behaviour, and product performance.

However, raw data requires cleaning, transformation, and analysis to generate meaningful insights.

You will build a data analysis pipeline using NumPy, Pandas, Matplotlib, and Seaborn to clean data,

perform analysis, visualize patterns, and derive actionable business insights.

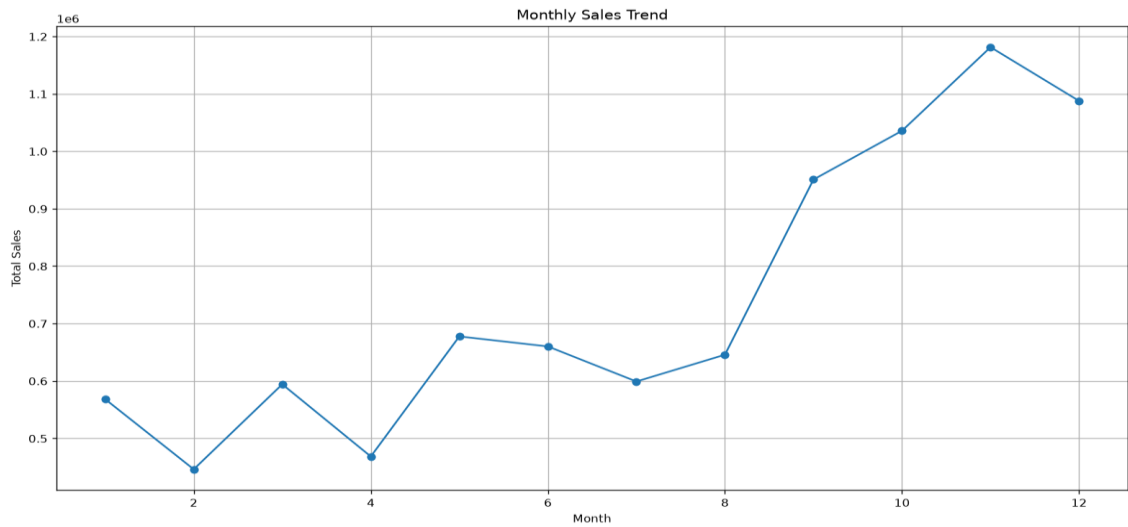
About the Dataset — Online Retail

1. **What it is:** A transactional dataset from a UK-based online gift retailer, covering **541,909 transactions from December 2010 to December 2011**, with 8 columns — InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country. Most buyers are wholesalers rather than individual consumers, and transactions span 37 countries (mainly the UK).
2. **Data quality & cleaning:** The raw data had ~135,000 rows with missing CustomerID, negative Quantity values (returns/cancellations), invalid UnitPrice entries, and duplicate rows. After cleaning, the dataset was reduced to roughly **400,000 valid transactions across 4,338 unique customers**, which formed the basis for the purchasing-behavior analysis (histograms, box/violin plots, heatmaps, and pairplots)

Task 7, 8 – Business Insights with visualizations

1. Line Chart – Best Sales Month

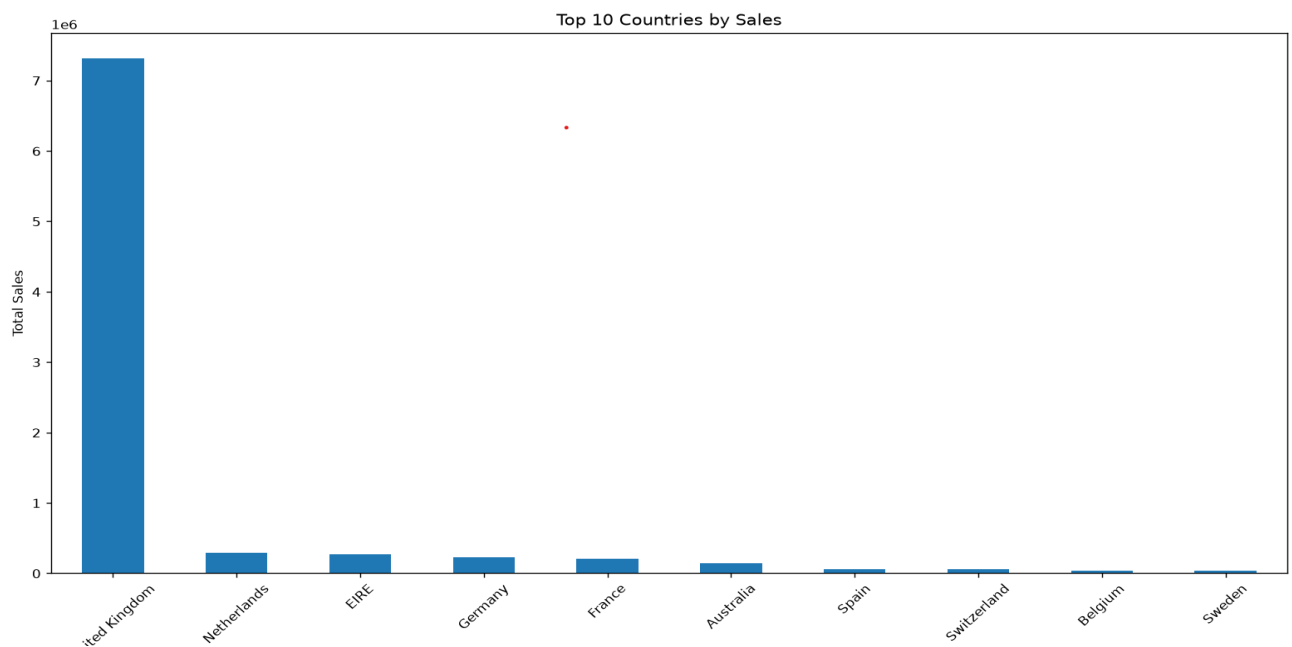
Business insight:



Sharp growth from Aug–Nov — sales nearly double, climbing from ~0.65M to a peak of ~1.18M in November (holiday season).

Slight dip in December after the November peak, while Jan–Jul stays relatively flat and volatile (~0.45M–0.68M)

2. Bar Chart – Top Countries



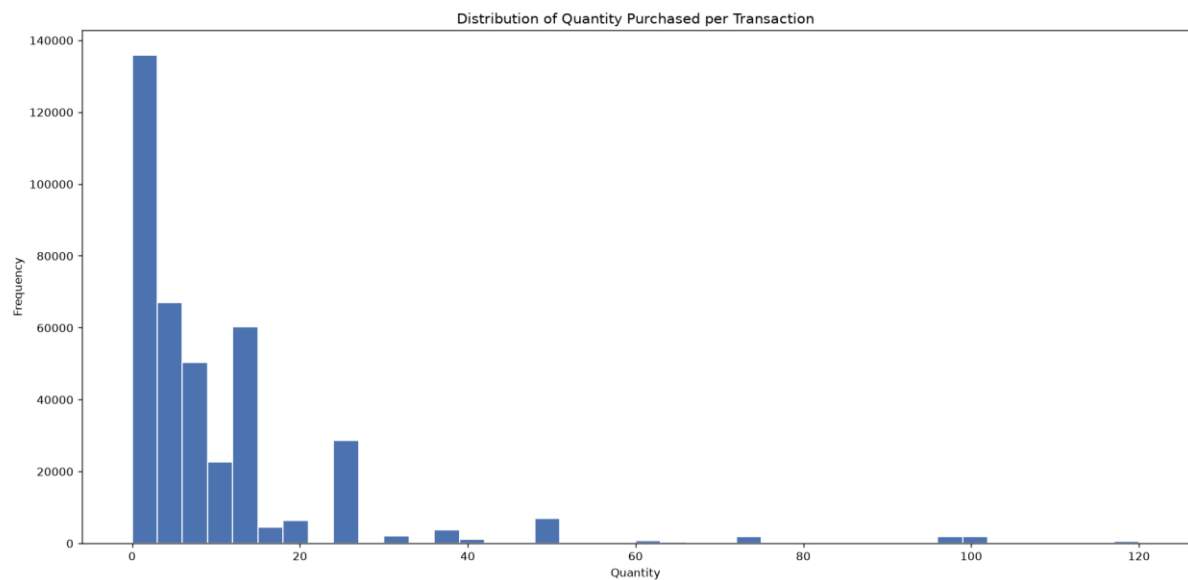
Business insights:

UK dominates overwhelmingly — total sales (~7.3M) are roughly 25x higher than the next country, confirming this is primarily a domestic UK retailer.

Rest of top 10 trail far behind — Netherlands, EIRE, and Germany form a distant second tier (~200–300K each), with remaining countries under 150K

Histogram:

Customer behaviour



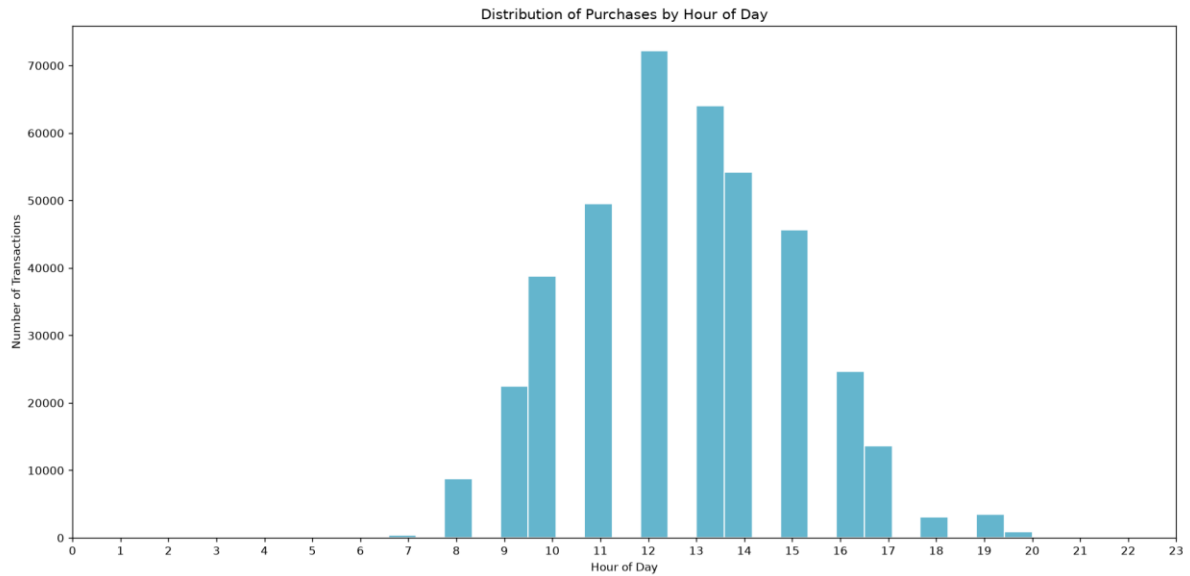
Business insight:

Right-skewed — most orders are small (median = 5 units), with a few bulk orders creating a long tail.

Zero-spike — from cancelled/returned quantities reset to 0 during cleaning, not real purchases.

Peak sales time

Histogram: Purchases by Hour of Day

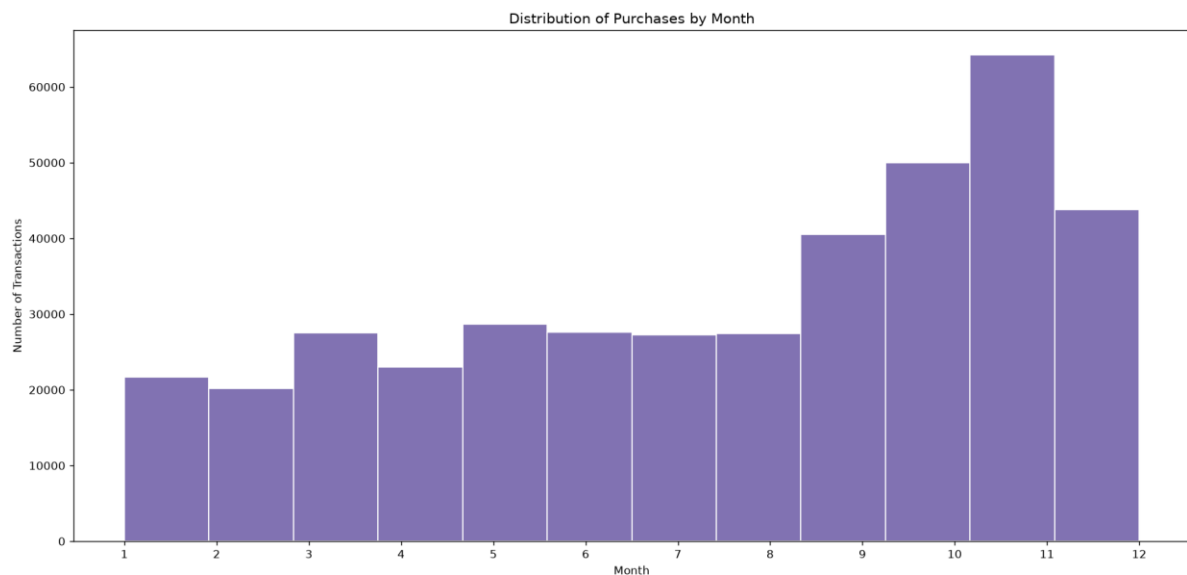


Business inference:

Peak activity midday — purchases peak sharply between 12–14 hrs, with strong volume from 10 AM–3 PM.

No purchases outside business hours — near-zero activity before 7 AM or after 8 PM, consistent with B2B/office ordering hours.

Histogram: Purchases by Month

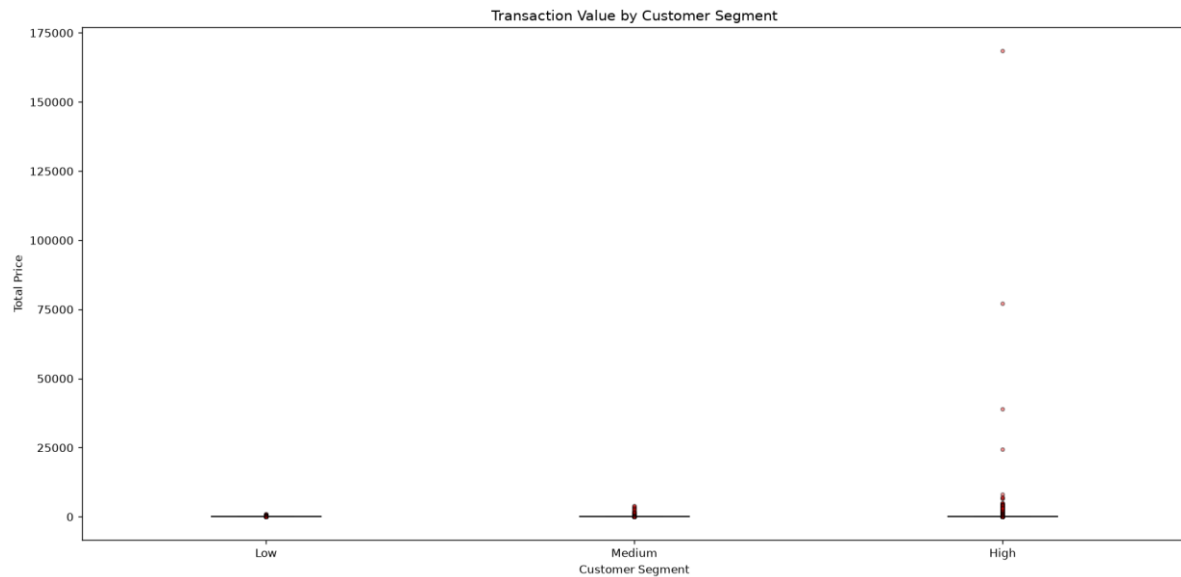


Seasonal peak in Q4 — transactions rise sharply from September through November (holiday/Christmas stocking), peaking in November.

Steady, lower baseline Jan–Aug — relatively flat activity for the first 8 months before the seasonal surge

4.High value customers

Box plot: Total Price per Transaction by Customer Segment



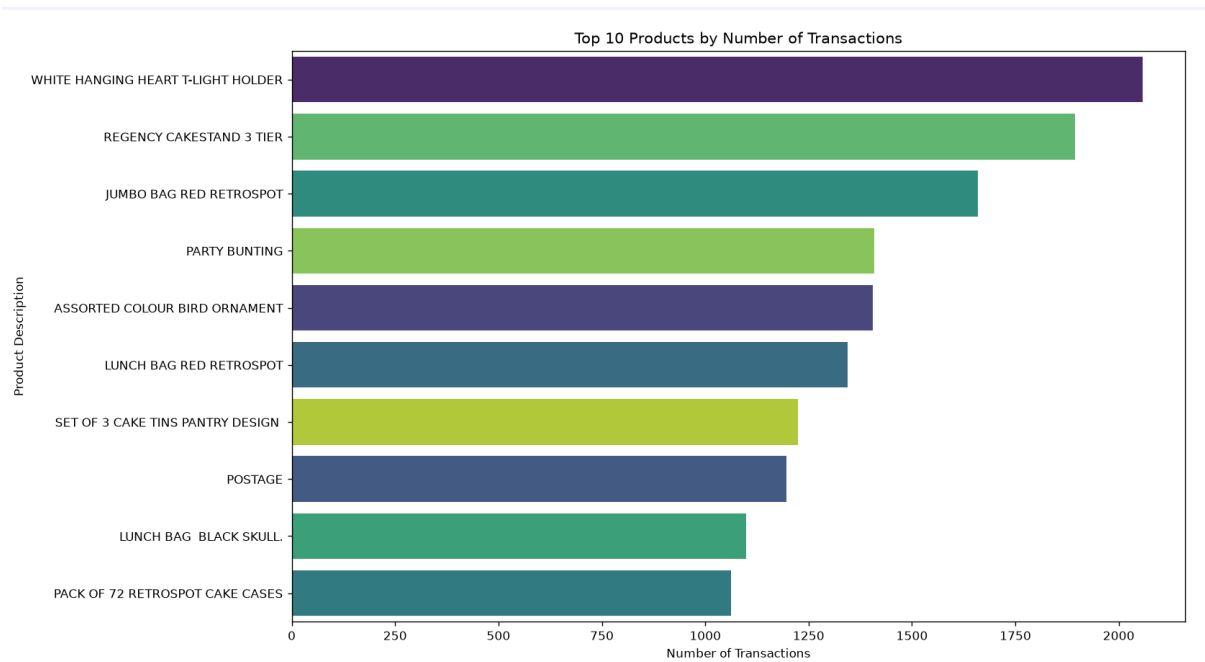
Business insights:

High segment has by far the widest spread and largest outliers (up to ~£168,000), confirming it drives the biggest purchases.

Low/Medium segments stay tightly clustered near zero, with only minor outliers

5. Top products

Count Plot – Top 10 Products by Number of Transactions

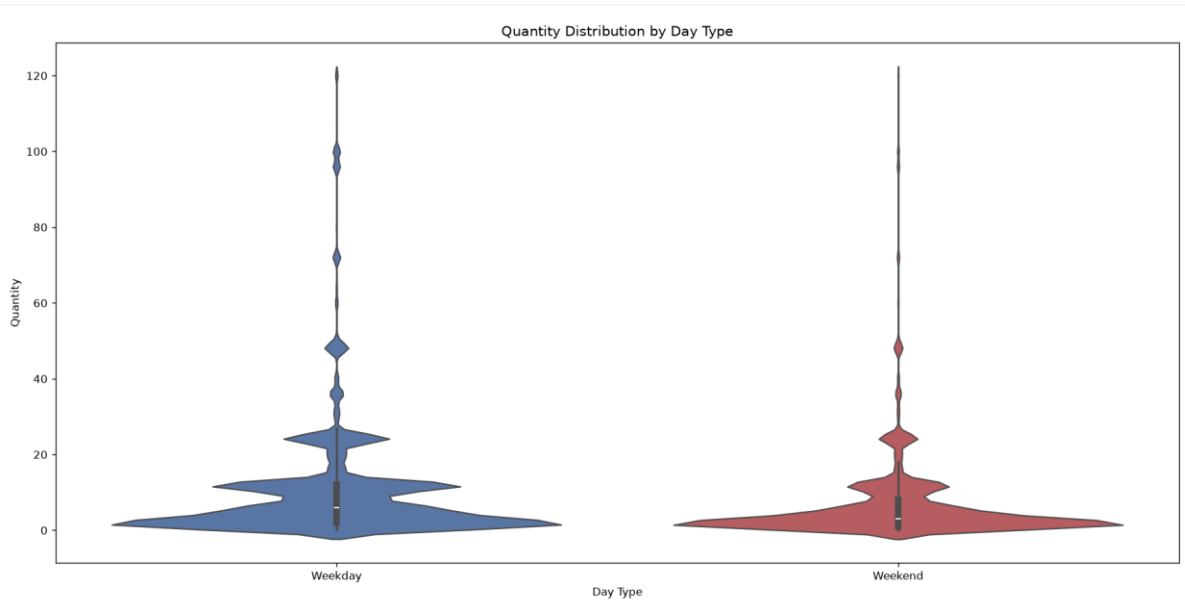


Business insights:

"White Hanging Heart T-Light Holder" leads with 2,000+ transactions, followed by "Regency Cakestand 3 Tier" and "Jumbo Bag Red Retrospot."

Gradual decline, no single dominant product — top 10 counts range narrowly (~1,000–2,000), showing demand is spread across several popular items rather than concentrated in one

6. Violin Plot – Quantity by Day Type using seaborn

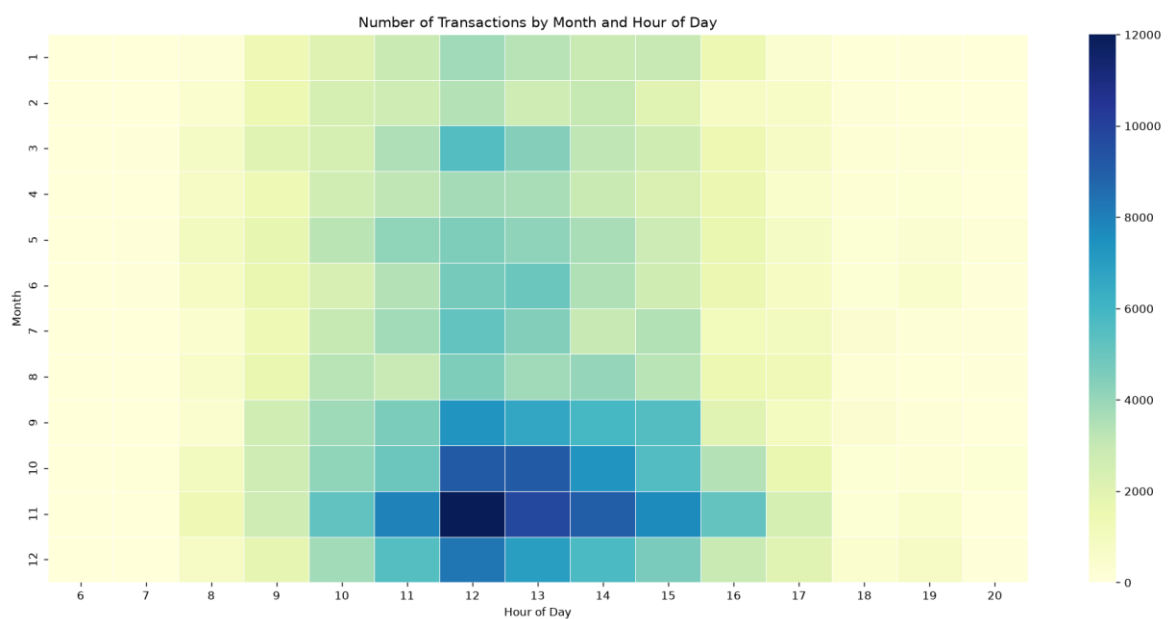


Business insights:

Nearly identical distributions — Weekday and Weekend quantities have the same shape, median (~5–6), and spread, so day type has little effect on order size.

Both are right-skewed with multiple bulges (common bulk sizes like 12, 24) and a thin tail of rare large orders.

7. Heatmap – Transaction Volume by Month vs Hour of Day (seaborn)

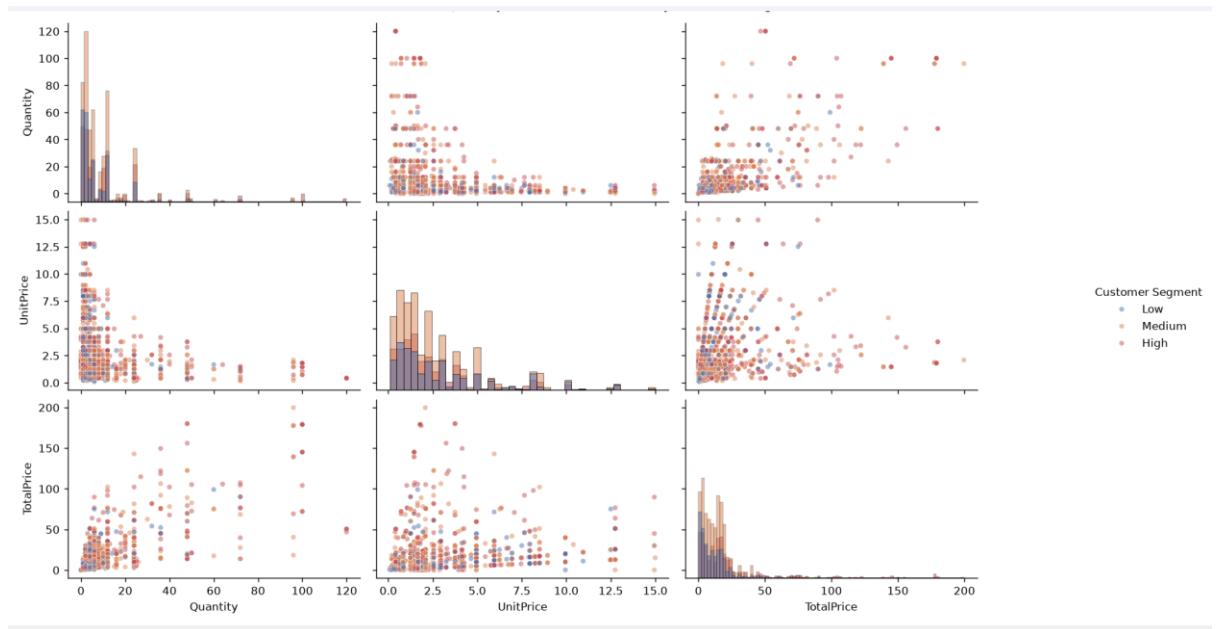


Business insights:

Peak activity: November around 12–1 PM — the darkest cell, confirming both the Q4 seasonal surge and the midday buying pattern overlap into one clear busiest window.

Consistent daily rhythm year-round — every month shows the same 10 AM–3 PM concentration, just scaling up in intensity from September to December.

8. Pair Plot – Quantity, UnitPrice, TotalPrice (colored by Customer Segment) using seaborn



Business insights:

TotalPrice rises with Quantity — a clear positive relationship, since TotalPrice is derived from $\text{Quantity} \times \text{UnitPrice}$.

All three variables are right-skewed with no strong Quantity–UnitPrice link, and High-segment (red) points spread further into higher-value regions, confirming they drive the biggest purchases